

WHAT MONTHLY MODELS MISS: OIL PRICE TIMING AND US CONSUMER ENERGY PRICE FORECASTING

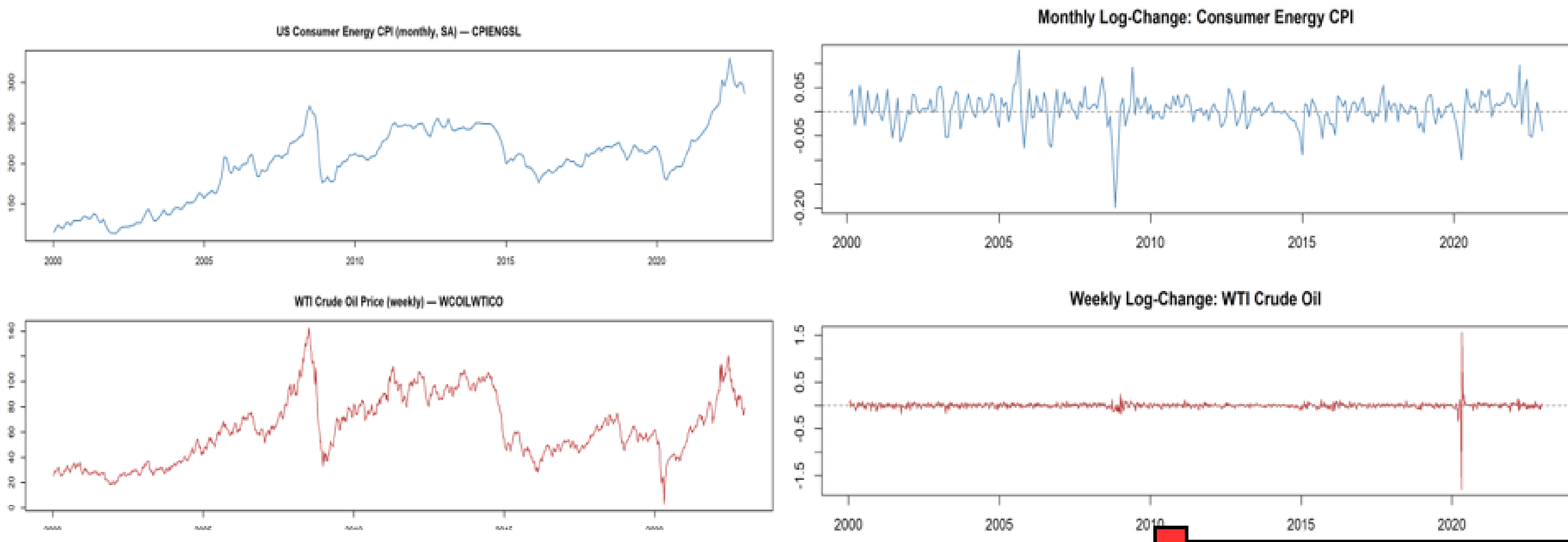
A mixed-frequency comparison of econometric, state-space, penalized, and machine-learning models, with historical backtesting, post-2022 external holdout, and forward scenario forecasts

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Situation

The Aggregation Problem in Energy Price Forecasted

- US consumer energy CPI is observed monthly, while WTI crude oil prices are available weekly.
- Standard monthly models typically aggregate weekly oil movements into a single monthly average before estimation.
- This pre-aggregation can hide useful timing information if some weeks matter more than others for consumer energy price transmission.
- The result is a potentially weaker forecast and a less informative view of how oil-price shocks reach consumers.

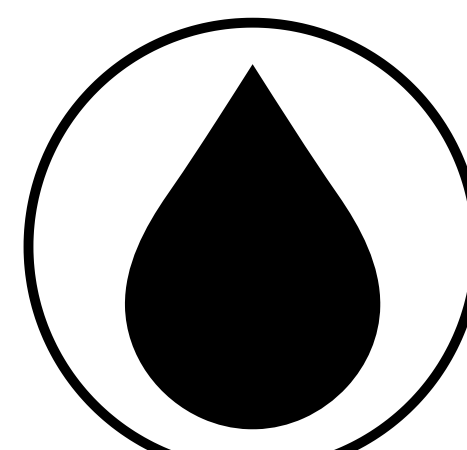


Task

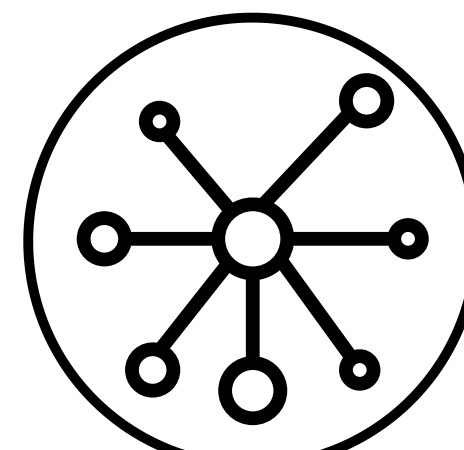
Build and Evaluate Better Forecasts

- Test whether preserving weekly oil-price timing improves forecasts of monthly CPI Energy relative to standard monthly aggregation.
- Compare a broad set of models under one common forecasting design: ARIMAX, MIDAS, U-MIDAS, CLM-SS, LASSO-MIDAS, Kernel U-MIDAS, XGBoost, and LSTM.
- Evaluate models using historical rolling-origin backtests and a genuinely later external holdout from 2023 to 2026.
- Assess not only forecast accuracy, but also directional usefulness and interpretability.

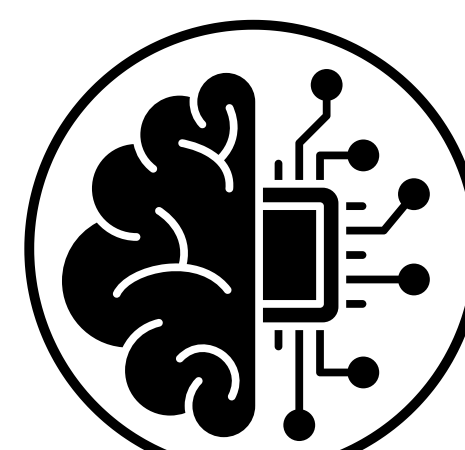
Forecasting Pipeline



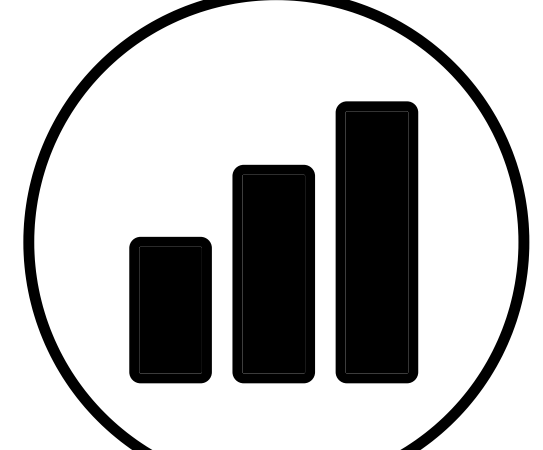
Weekly
WTI



MIDAS
Weighting



Forecasts
Models



Monthly
CPI-Energy
Forecast



Evaluation
(RMSE / DA)

MAIN CONTRIBUTION

~16% Lower RMSE on average in historical backtests using mixed frequency models with weekly oil data

81.2% Directional accuracy in the post-2022 external holdout

7/9 Models beat the monthly benchmark out-of-sample

$$y_t = \alpha + \beta \sum_{k=0}^K w(k; \theta) x_{t-k/m}^{(m)} + \varepsilon_t$$

Action

Mixed-Frequency Forecasting Design

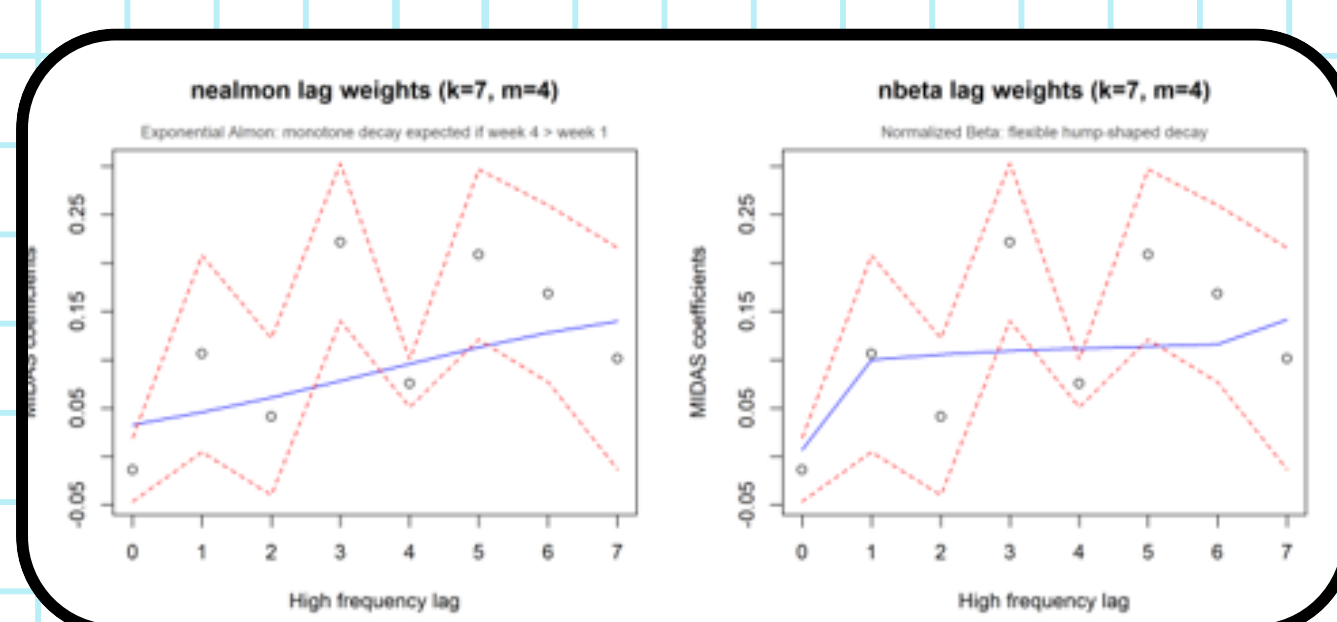
- Target variable: monthly log-change in US Consumer Energy CPI.
- Main predictor: weekly log-change in WTI crude oil prices.
- Sample: 2000-2022 for model development, 2015-2022 for historical rolling-origin evaluation, and 2023-2026 as an external holdout.
- Models include benchmark monthly, mixed-frequency, penalized, state-space, and machine-learning approaches.
- The objective is to compare not only predictive accuracy, but also whether the oil-price signal remains interpretable.

Modeling Framework (MIDAS Regression)

$$y_t = \alpha + \beta x_t^{(m)} + \sum_k w_k(\theta) x_{t-k}^{(w)} + \varepsilon_t$$

- y_t : Monthly CPI Energy log-change
- $x_t^{(m)}$: Monthly controls / autoregressive terms
- $x_{t-k}^{(w)}$: Weekly WTI information
- $w_k(\theta)$: MIDAS lag weight

MIDAS Models estimates how much each weekly oil-price lag contributes to the monthly CPI energy forecasts



Result

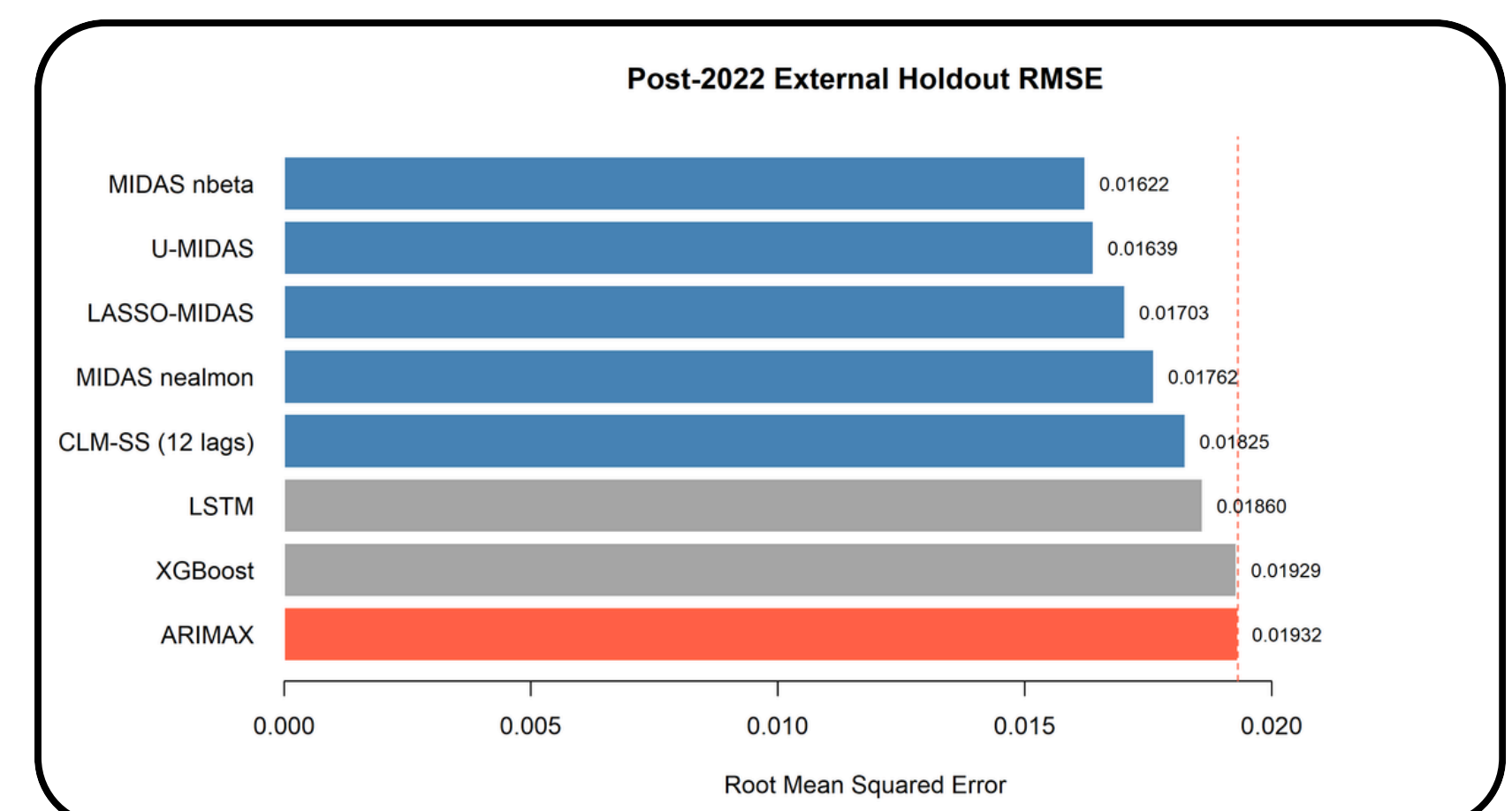
Stronger Forecasting Evidence with Weekly Oil Info

Historical OOS Performance (2015-2022)

Model	RMSE	Directional Accuracy
ARIMAX	0.0276	64.60%
MIDAS nealmon	0.0341	81.20%
MIDAS nbeta	0.0291	78.10%
U-MIDAS	0.0271	78.10%
CLM-SS (12 lags)	0.0275	76.00%
XGBoost	0.0217	65.80%
LSTM	0.0251	63.20%

Post-2022 External Holdout (2023-2026)

Model	RMSE	Improvement vs ARIMAX
MIDAS nbeta	0.01622	16.00%
U-MIDAS	0.0164	15.10%
LASSO-MIDAS	0.01703	11.90%
MIDAS nealmon	0.01762	8.80%
CLM-SS (12 lags)	0.01825	5.50%
LSTM	0.0186	3.70%
XGBoost	0.01929	0.10%
ARIMAX	0.01932	baseline



Implications

- Weekly oil-price timing contains useful information for forecasting monthly consumer energy inflation.
- Mixed-frequency models can improve both forecast quality and the interpretation of transmission timing.
- For practitioners, the result suggests that pre-aggregation may discard relevant information too early.
- For research, the findings support using mixed-frequency designs when timing itself may carry economic meaning.

Limitations

- The application focuses on one domain: US consumer energy prices and oil.
- Model rankings are not identical across all evaluation windows.
- Some flexible models, especially deep learning, are constrained by the relatively small monthly sample.
- MAPE is reported cautiously because monthly log-changes can be close to zero, which inflates percentage-error measures.

Future Work

- Extend the framework to other energy-sensitive prices and other commodities.
- Test richer mixed-frequency structures using daily or real-time vintage data.
- Explore regime-switching and nonlinear mixed-frequency models in greater depth.
- Develop the approach further into a reusable forecasting toolkit or package.